

Running the Model in Front of the Examiner

A step-by-step guide for the defence

IEEE 13-node feeder, DIgSILENT PowerFactory 2021 — Jhala Nath Kafle (081MSPSE009)

The short version. Open PowerFactory, activate the project **IEEE13 DSDR Fuse Coordination** and the study case **Study with Substation Transformer**. Then show three things by hand: (1) the load flow with the DG out and in — the current through R2 reverses; (2) a short circuit at node 633 — the fuse carries more current than the recloser; (3) the tripping times of the relays and the fuse. That takes about five minutes and proves the model is real. The scripts are the backup, not the main show.

1. Before the defence (do this the day before)

1. **Use the same laptop** you built the model on. The project is stored in PowerFactory's own database on this machine, not in the project folder.
2. **Check the licence.** Open PowerFactory once where you will present. If the licence needs a network or a dongle, make sure it works there.
3. **Rehearse every step of Section 2 once**, and write down the numbers you see next to the numbers in Section 5. The menu names below are those of PowerFactory 2021; if a button is somewhere else on your screen, note where.
4. **Leave the model in the DSDR state.** It is left that way after the last run of the study.
5. **Keep a backup:** in PowerFactory, right-click the project → *Export*, and save the .pfd file on a USB stick.
6. **Have these open and minimised:** PowerFactory with the project active; the report PDF; and the figures folder, results\figures, inside DSDR_Study.

2. Showing the model by hand in PowerFactory (recommended)

Step 1 — Open and activate (30 seconds)

1. Start **PowerFactory 2021**.
2. Open the Data Manager, find the project **IEEE13 DSDR Fuse Coordination** under your user, right-click → **Activate**.
3. In the study-case list at the top, choose **Study with Substation Transformer**.
4. The single-line diagram of the IEEE 13-node feeder appears.

Say: "This is the IEEE 13-node feeder at 4.16 kV, fed from a 115 kV grid through a 5 MVA transformer. R1 is here at the feeder head, R2 is on line 632 to 671, the fuses are on the laterals, and the DG is at node 692."

Step 2 — Load flow without the DG (1 minute)

1. Right-click the **Synchronous Machine** at node 692 → set it **Out of Service**.
2. Click **Calculate Load Flow**. Select *AC Load Flow, unbalanced, 3-phase (ABC)* and **Execute**.
3. Point at the result boxes of line 650–632 and of line 632–671.

Say: "Without DG the feeder head carries about 588 amperes and the branch of R2 about 470 amperes, flowing from the grid towards the loads."

Step 3 — Load flow with the DG (1 minute)

1. Put the Synchronous Machine back **In Service**.
2. **Calculate Load Flow** again.
3. Point at line 632–671: the current is now about 252 amperes and the power flows *towards* 632.

Say: “With the DG the current through R2 falls to about 252 amperes and its direction reverses. This is why R2 needs a second, reverse setting: 1.25 times 252 gives the reverse pickup of 315 amperes.”

Step 4 — Short circuit (1 to 2 minutes)

1. Right-click the busbar **633** → **Calculate** → **Short-Circuit**.
2. Method: **complete**. Fault type: **3-phase short-circuit**. Fault impedance 0. **Execute**.
3. Point at the currents in the fuse F633, in R1 (line 650–632) and in R2 (line 632–671).

Say: “For a three-phase fault at 633 with the DG, the fuse carries about 5.4 kiloamperes, but R1 carries only about 4.0 and R2 about 1.5 in reverse. The fuse sees the grid and the DG current together; each recloser sees only its share. That is the whole problem.”

Step 5 — Tripping times (1 minute)

With the short circuit of Step 4 still calculated:

1. Open the relay **R1 Fast** (double-click it in the cubicle, or find it in the Data Manager) and look at its **tripping time**. Do the same for **R2 Rev Fast** and for the fuse **F633**.
2. Or open a **time-overcurrent plot** that contains these devices: the vertical line is the fault current, and the plot shows where it meets each curve.

Say: “R2 trips on its reverse group in about 0.06 seconds and R1 in about 0.11 seconds. The fuse would start to melt at about 0.15 seconds. So both reclosers trip first and the fuse is saved.”

Step 6 — If asked for the EMT simulation

The EMT run needs the scripted reclosing events, so it is quicker to show the saved result: Figure 5.14 of the report, or the picture file

DSDR_Study\results\figures\Fig10_EMT_LL_684.png

If the examiner insists on a live run, use Section 3, command (d); it runs the simulation six times and takes a few minutes.

3. Running the scripts (backup, or if asked “show me the code running”)

Close PowerFactory first. The scripts start their own PowerFactory in the background, and that cannot start while the PowerFactory window is open.

Open a Command Prompt and go to the study folder:

```
cd /d "C:\Users\Rabin\Desktop\Digsilent_project - TWIST AND TURN\DSDR_Study"
```

```
set PY="C:\Users\Rabin\AppData\Local\Programs\Python\Python39\python.exe"
```

	Command	What it does and what appears
(a)	%PY% step2_studies.py	Load flows and all short circuits in PowerFactory. Prints the currents through R1 and R2 for each fault.
(b)	%PY% step3_design_and_evaluate.py	No PowerFactory needed; a few seconds. Prints Table II, the settings, and the cell counts: 35 and 39 without DG, 24 with DG, 38 with the DSDR.
(c)	%PY% step4_apply_settings.py dsdr	Writes the DSDR settings into the relays and fuses and compares PowerFactory's tripping times with the calculated ones.
(d)	%PY% step5_fig10_emt.py	The EMT simulation, DG out and DG in. Prints the event times and saves the figure.

The safest script to run live is (b). It needs no PowerFactory, finishes in seconds and prints the headline results. The full output is also saved in `results\summary.txt`, which you can simply open.

Do not run `run_all.py` or `step1_build_model.py` at the defence. They rebuild the model and all outputs from scratch, which takes several minutes and is not needed to show a result.

4. A third way: the script inside PowerFactory

The project contains a Python script object named **Final(1)**. With the project active, find it in the Data Manager (under the study case or the project's *Scripts*), right-click → **Execute**. It runs the load flows and the short-circuit database and prints a report in the **Output Window**, with the current through R1 and R2 and its direction, DG out and DG in. Rehearse it once; it takes a few minutes.

5. Numbers you should see

Load flow	DG out	DG in
Feeder head, line 650–632 (R1)	587.7 A	284.9 A
Line 632–671 (R2)	470.0 A, forward	252.2 A, reverse

Bolted three-phase fault	DG out	DG in
Fault current at 632	4.74 kA	6.53 kA
Fault current at 633	4.10 kA	5.43 kA

Fault at 633, DG in, DSDR	Current	Device	Time
R2, reverse group	1500 A	R2 Rev Fast	0.060 s
R1	3970 A	R1 Fast	0.106 s
Fuse F633 (400E)	5426 A	F633	melts at 0.149 s

Small differences in the last digit are normal. If a number is very different, check the two things in Section 6 first.

6. If something goes wrong

What you see	What to do
The numbers are much too high (for example 6 to 8 kA at 632 without DG)	The wrong study case is active. Choose Study with Substation Transformer .
R2 current is 470 A although you expect 252 A	The DG is out of service. Put the Synchronous Machine in service and run the load flow again.
Load flow does not converge	Press <i>Reset Calculation</i> , check that nothing was switched off by mistake, run again.
“Short-circuit calculation not possible”	You chose a phase the node does not have (645 and 646 have b–c only, 684 has a–c, 611 has c, 652 has a). Choose node 633, 632 or 671.
A script says PowerFactory could not start	PowerFactory’s window is still open. Close it and run the command again.
A script stops with an error	Do not debug in front of the panel. Open <code>results\summary.txt</code> and the saved figures, and say the results were produced by the same script.
PowerFactory will not open at all	Show the report figures and <code>results\summary.txt</code> ; offer to run it afterwards.

7. What the examiner may ask while it runs

Which calculation method is this? Unbalanced, three-phase load flow; short circuit by the *complete* method, which starts from the load-flow state.

Why this study case and not the other one? This one includes the 115/4.16 kV, 5 MVA substation transformer. With it the fault levels match the IEEE short-circuit benchmark within about 2 per cent. The stiff-grid case gives fault levels that are too high.

Where are the two settings of R2? The CDG34 relay of the library is not directional. So R2 has separate relay units: *R2 Fast* and *R2 Delayed* for the forward group, *R2 Rev Fast* and *R2 Rev Delayed* for the reverse group. The direction is assigned from the fault location. I state this as a limitation.

What do the Python scripts do that PowerFactory does not? PowerFactory does the load flow, the short circuits and the EMT simulation. The scripts run those studies for all nodes and fault types, apply the method’s equations to get the settings, and classify the 39 cells. Doing 273 short circuits by hand would not be practical.

Can you change the DG size and show the effect? Open the Synchronous Machine, change its active power, run the load flow: the current through R2 changes. The full study at seven penetration levels is in Section 5.16 of the report.